

- 5 (a) displacement & direction of energy travel normal to one another ... B1 [1]
- (b) (i) phase angle of 60° correct .. (need to see $1\frac{1}{2}$ wavelengths) B1
lags behind T_1 B1 [2]
- (ii) waves must be in same place (at same time) B1
resultant displacement = sum of individual displacements B1 [2]
- (iii) 1. $-\frac{1}{2}A$ B1
2. $\frac{1}{2}A$ (allow e.c.f.) B1
3. zero (allow e.c.f.) B1 [3]